

Boomerang mobile counter shooter detection system

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ABSTRACT

Boomerang is an acoustic system installed on military vehicles that is designed to detect relative shooter azimuth/range/elevation from incoming small arms fire. It performs passive acoustic detection and computer-based signal processing. Aural and visual alerts are used to 1) inform vehicle occupants that a bullet has passed within close proximity to the vehicle and 2) indicate the position of the shooter relative to the vehicle's direction of travel. Boomerang operates when the vehicle is stationary or moving (no motion compensation) using a single, compact, mast-mounted array of microphones. The system is calibrated to detect infantry small arms. This calibration, however, does not preclude the system from detecting larger and smaller supersonic rounds. In this paper, we discuss the design, development, testing and production of 50 Boomerang I systems over a 65-day period in late 2003/early 2004. These systems were deployed during Operation Iraqi Freedom II with Marine and Army units. Feedback from operational units identified specific deficiencies and desired improvements that were incorporated into a system re-design effort, Boomerang II. The Boomerang II system has been extensively tested for performance and environmental fitness regarding RF compatibility with tactical radios (SINCGARS), heat, cold, shock and vibration. The US Army Aberdeen Proving Grounds conducted successful RF compatibility tests, road tests and 'live fire' testing. Summary results of open field 'live fire' static tests are presented as well as performance results on a remote-controlled HMMWV operating at 45 MPH.

Keywords: Sniper, counter-sniper, gunfire, acoustic, shock wave, muzzle blast, bullet, Boomerang

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1. INTRODUCTION

In the Fall of 2003, the US was incurring a high number of troop casualties while conducting military operations in Operation Iraqi Freedom I (OIF I). A large number of incidents occurred during routine logistics convoy operations. At the time, insurgents would hide along convoy routes approximately 100 meters from the road. When the convoy arrived, the insurgents would fire their AK-47's in short bursts from the safety of buildings, behind walls, etc. In many cases, these shots were not heard due to the high noise environment of the HMMWV combined with the sound deadening effect of military helmets. A typical marine corporal response when asked "When did you know you had been fired upon?," would result in three answers: 1) the vehicle window got 'spidered', 2) bullet holes were observed in the vehicle upon arrival at our destination or 3) someone took a hit (casualty).

Several "commercial-off-the-shelf," (COTS) counter sniper systems were considered as a potential solution, however, these systems performed poorly in urban environments and their unit cost was very high. The Defense Advanced Research Project Agency (DARPA) initiated a rapid response effort, termed Boomerang, to develop and field 50 shooter detection systems in 60 days. BBN Technologies had developed a stationary proof-of-concept acoustic counter sniper system called 'Bullet Ears' during a mid-90's DARPA Counter Sniper program. Boomerang was the accelerated development of this early system with the additional requirement that it operate on a moving vehicle (e.g. HMMWV), have a low unit production cost, perform reliably in noisy and reverberant urban environments and be easy to install...produced and delivered in 60 days. BBN designed, prototyped, tested and produced 50 Boomerang I systems in the 65-day period from 18 November 2003 to 21 January 2004.

1. SYSTEM DESCRIPTION

1.1. Boomerang I – First article design

Given the time constraints and time of year (Holidays), Boomerang I was designed primarily using Commercial-Off-The-Shelf (COTS) components. In addition, vendors and products had to be selected based on their ability to deliver materials in the quantities and time frame required to meet the desired delivery date. While BBN had significant experience and corporate memory in developing a prototype counter sniper system ('Bullet Ears') based on a mid-90's DARPA Counter Sniper program¹, there was no relevant hardware starting point so the entire system architecture had to be designed. In developing the system, special consideration was given to minimizing hardware cost, keeping the overall system simple to operate and install and ensuring that the system would operate reliably over a wide range of conditions.

1.1.1. Goal

The initial goals of the Boomerang I system were the following:

- Find relative shooter direction to ± 15 deg ('hours of a clock') within 1 second of shot
- Missed shot detections $< 1/1000$
- Shooter direction errors greater than ± 15 deg $< 1/500$
- Shot miss distance (CPA): 1 meter - 30 meters
- Visually distinctive external sensor array (can become a decoy)
- Voice announcement and LED visual "rosette," display of shooter
- Shooter azimuth clockwise from ahead looking down
- Array diameter < 1 m (desired = 0.5 m) and with full azimuth view
- Vehicle power only, operator controls: on/off, volume, brightness
- Urban operations, low buildings, shooter range 50 m - 150 m
- Operates at up to 60 mph on rough terrain or highways

- Withstands sand, pebbles, rain and light foliage impacts
- Operates at temperatures from 0 - 50 degree Celsius
- No shooter elevation indication
- Field installation
- Field replaceable microphone array and electronics box

1.1.1. System architecture and components

The overall system architecture of Boomerang is illustrated in the block diagram shown in Fig. 1. The system consists of three major components:

- 1) Acoustic sensor array mounted on top of a mast that is attached to the rear of the vehicle.
- 1) Signal processing unit in an electronics enclosure, using 24 VDC vehicle power, that is located under the rear, passenger seat.
- 1) User interface consisting of a display/loudspeaker/GPS located on the vehicle dashboard.

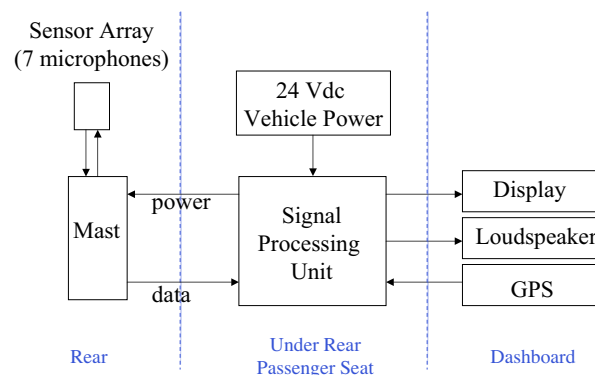


Fig. 1. Boomerang system block diagram.

The installed system on a hardback HMMWV is depicted in Figure 2. The cable from the mast to SPU has a quick disconnect at a bulkhead connection located at the base of the mast to permit removal of the sensor array. Also shown is a WiFi antenna (802.11) that is magnetic-mounted to the roof. The purpose of the WiFi and GPS were to permit future system improvements. The WiFi system was also used during system testing as a means to transmit data in real-time to a remote display and computer.

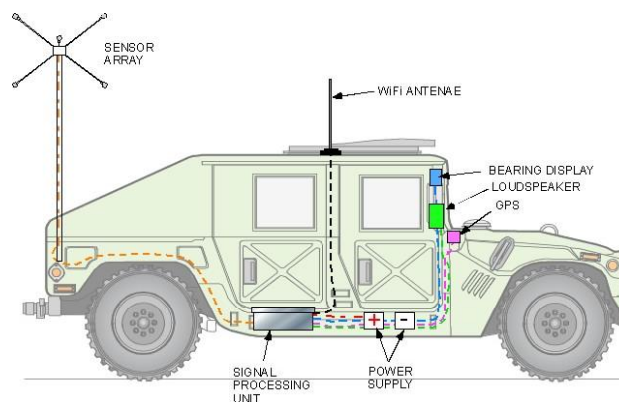


Fig. 2. Boomerang I system installed on hardback HMMWV.

The Boomerang I array (Fig. 3) consisted of seven microphones attached to 'spines' in a 1-meter diameter volumetric array configuration. BBN used previously developed ballistic and shock wave models^{1,2,3} to guide the compact array

design. While a conformal, distributed sensor array would have provided a large acoustic baseline, the requirement for ease of installation led to an early design decision to fix the sensors in a known spatial geometry in order to control relative sensor positioning. A custom signal conditioning and power supply board was designed and installed inside the hub at the top of the mast. The microphone 'spines' were threaded into the hub where each individual microphone was connected to the electronics board inside the hub. The hub was attached to the mast using rubber isolation mounts. The microphones did not require calibration; therefore, damaged or defective 'spines' could be replaced without concern for special matched sets and tedious system re-calibration.

The Signal Processing Unit (SPU) consisted of a compact 650 MHz Intel Celeron processor motherboard, analog to digital PC-100 daughter board, power conditioning and distribution boards, audio power amplifier module, and a custom analog Time-Delay-of-Arrival (TDOA) board. A FLASH memory card, installed into a second PC-100 daughter board, was used to store 100 shot data files (time series data) that could be downloaded for subsequent post-processing and analysis. During field tests, these shot data files became an extensive database for evaluating the performance of software changes and algorithm improvements. The WiFi interface was provided by a PCMCIA card that occupied the second slot in the second PC-100 daughter board. The processor motherboard was mounted on a bedplate that was in shock mounted inside the SPU enclosure. The SPU enclosure was water-resistant and provided connections for the mast cable, power supply, display, and WiFi antenna. System operation was enabled via a single Power ON/OFF switch, also mounted on the SPU enclosure.

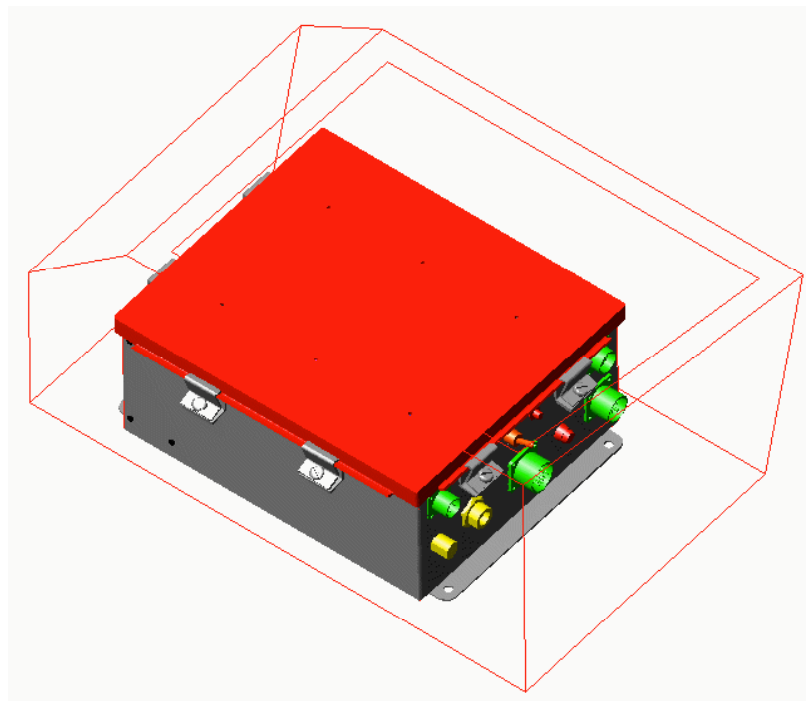


Fig. 3. Boomerang I sensor array and Signal Processing Unit (SPU).

The alerting mechanism, shown in Fig. 4, consisted of a COTS display that was originally intended for RF direction finding applications. The display provided a rosette of sixteen red light emitting diodes (LEDs) that were used to

indicate the relative shooter direction in compass fashion. An aural alert (“incoming x o’clock!,,) is announced via a COTS external audio speaker.



Fig. 4. Boomerang I COTS display.

1.1.2. Installation

The Boomerang system is easily installed on several variants of HMMWV's. Frequently, HMMWV's are equipped with a SINCGARS radio and antenna and the Boomerang mast must be installed on the opposite side of the vehicle from the SINCGARS antenna. The provided vehicle cables, between the signal processing unit and the mast, are long enough to facilitate installing the mast on either the passenger or the driver's side of the vehicle. A reinforcing fender brace, as shown in Fig. 5, is required between the HMMWV fender and the vehicle chassis to provide adequate structural support for the mast.

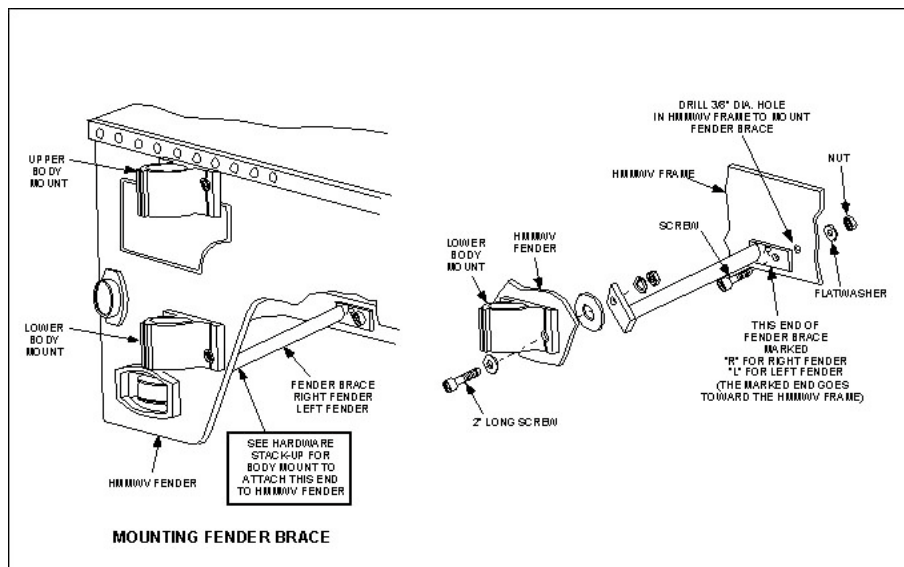


Fig. 5. Boomerang mast mounting brackets and fender brace.

The Boomerang display is typically mounted on the plenum between the two windshields with sheet metal screws as shown in Fig. 6. The GPS receiver is mounted nearby with self-adhesive Velcro strips. A single multi-conductor cable is run along the outside of the engine cover from the display unit to the Signal Processing Unit, located under the passenger rear seat, and is held in place with cable clamps. Two 1-1/2" holes in the compartment under the rear passenger seat and in the battery compartment are required to route cables to the Signal Processing Unit.

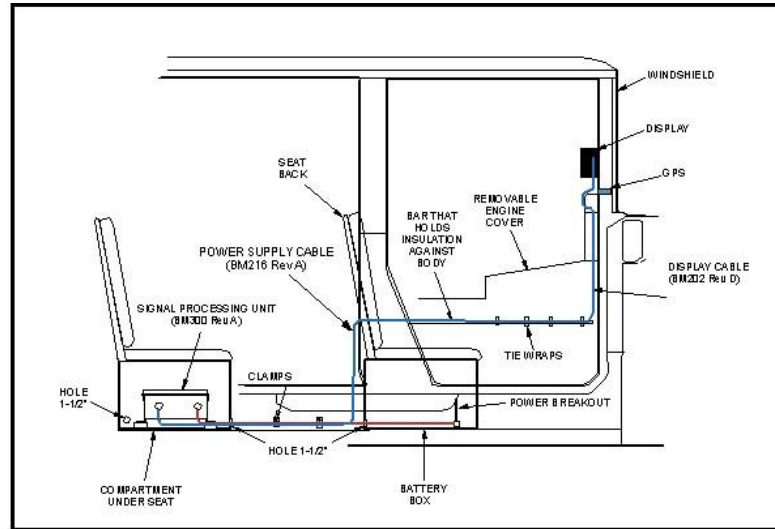


Fig. 6. Display cable routing.

Boomerang I systems were installed on Marine vehicles in Kuwait in March 2004 as shown in Fig. 7. These systems provided rapid feedback from the Marines on system performance and desired improvements. System installation can be accomplished in approximately four man-hours. A recently installed Boomerang II system is shown in Fig. 8.



Fig. 7. Boomerang I systems installed in Kuwait (2004).



Fig. 8. Re-designed Boomerang II system installed at Camp LeJeune (2005)

1.1.3. User feedback

One of the primary motivations for the rapid deployment of Boomerang was to deliver systems to operational units who could provide feedback on system performance, problems and recommended improvements. After Boomerang I was delivered and installed with operational units, it became difficult to obtain feedback on system performance. Several problems were reported.

Problem 1: Electrical short in battery compartment that damaged vehicle electrical cabling.

This problem was the result of a separate power conditioner that was installed between the Boomerang SPU and the vehicle power. The initial shipment of power conditioners from the vendor came with mounting flanges that provide a means to secure the component to the wall inside the battery compartment. A second shipment arrived without the mounting flanges. These were installed by wedging the power conditioner between the compartment wall and battery along with Velcro adhesive strips. Due to vehicle vibration and shock, these units dislodged and shorted the battery.

Problem 2: System false alerted.

This problem was eventually attributed to an incompatibility with high power RF systems, specifically, SINCGARS radio. During the initial testing of Boomerang I, there was no opportunity to conduct tests with tactical radios and it was recommended that the Boomerang I system be installed on vehicles without tactical radios. In some instances, Boomerang I was installed on HMMWV's that were tactical command vehicles having multiple high power radio systems.

Problem 3: Hub mount failure.

The hub mount was designed for compressive loads. Under tension and high temperatures, the mount failed.

1.2. Boomerang II – Re-designed and enhanced

Although the Boomerang I problems were disappointing, there was sufficient progress and performance to warrant a re-design effort. In addition, a number of system improvements were suggested by the operational user community.

1.2.1. New requirements

Additional requirements of the Boomerang II system were the following:

- Determine range and elevation
- Shield for RF susceptibility and emissions
- Enhance display unit to include alphanumeric display for UTC coordinates (from GPS), range and elevation

1.2.2. Component redesign

The basic configuration of Boomerang II remained essentially identical to Boomerang I as shown in Fig. 9.

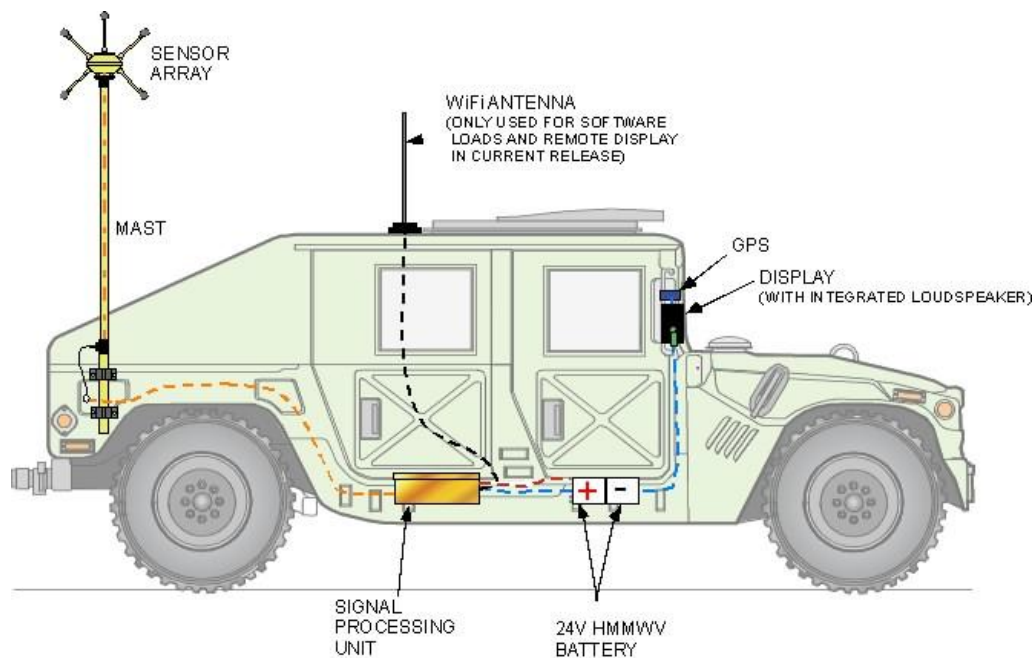


Fig. 9. Re-designed Boomerang II system installed on hardback HMMWV.

1.2.2.1. Software

The existing Boomerang software algorithms were augmented to determine range and elevation. Modifications were also made to support the updated Sensor and Display components described in following sections.

1.2.2.2. Electromagnetic shielding

All components of the redesign were surveyed and modified, as necessary, in order to minimize RF interference emitted by and received by Boomerang. Modifications included inserting conductive gaskets between mating parts, applying conductive coatings to enclosures, adding additional power supply filtering, using ferrite dampers, and upgrading existing shielded wires and connectors.

1.2.2.3. Sensor array

The original sensor array design required that low-level analog microphone signals be conducted along half the length of the vehicle using twisted wire pairs. Even with shielded twisted pairs, nearby (frequently co-located) radio antenna cabling easily introduced sufficient RF interference onto the sensor signals to render the signals unusable. A decision was therefore made to convert the analog signals into digital data and transmit the resulting high-level, noise-resistant digital signals to the SPU.

This decision required a radical redesign of the hub electronics. The original custom signal conditioning and power electronics board was modified to drive a new, additional custom board which performs the digital conversion and transmission tasks of the modified Sensor Array ("DigiHub,,). DigiHub uses Ethernet to provide a standardized, low-noise, high-bandwidth digital communications channel that interfaces directly into a pre-existing Ethernet port on the SPU. DigiHub A/D conversion is performed by four high-fidelity consumer stereo analog-to-digital converter (ADC) chips, with the surplus eighth channel used to monitor hub temperature. A field-programmable gate array (FPGA) is used to configure the ADC and Ethernet controller chips, muster the ADC digital data, format the digital data for transmission, and drive the formatted data out through the Ethernet controller chip. DigiHub power is provided using a modified implementation of the Power Over Ethernet (POE) standard. The entire electronics implementation fits onto two 2-sided surface-mount printed circuit boards (PCBs), approximately 3, in diameter, which are housed in a redesigned, thermally dissipative, electromagnetically shielded hub.

The user community also expressed a desire for a smaller array size. This was accommodated using a small, uniform length sensor 'spine' resulting in a 0.5-meter diameter array. The 'spine' connection to the hub was replaced with a quick-disconnect providing easier replacement of damaged sensors.

Finally, to address and solve the issue of hub-mount failure, the hub-to-mast mount was re-designed to be a captive configuration. The final design, shown in Fig. 10, is a compact, mechanically secure array that reduces RF interference from co-located, high-powered SINCGARS radios to negligible levels.



Fig. 10. Boomerang II sensor array with DigiHub and captive mast mount.

The time series data shown in Fig. 11 is an example of the type of SINGARS RF interference experienced in the Boomerang I system that would cause system false alerts due to inadequate electrical shielding.

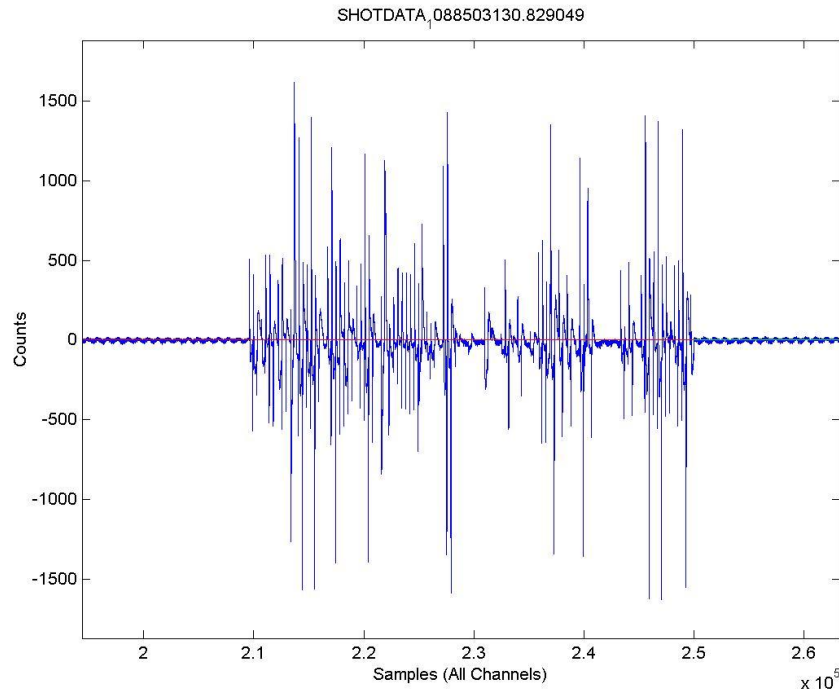


Fig. 11. Typical SINGARS RF interference on Boomerang I

The time series data shown in Fig. 12 is an example of a shot detected under similar high power SINGARS radio operation using the re-designed DigiHub. The shock wave (first peak) and the muzzle blast (second, lower peak) are clearly identifiable while being exposed to high power SINGARS radio transmissions. This system re-design also resulted in significantly reduced system electrical noise levels.

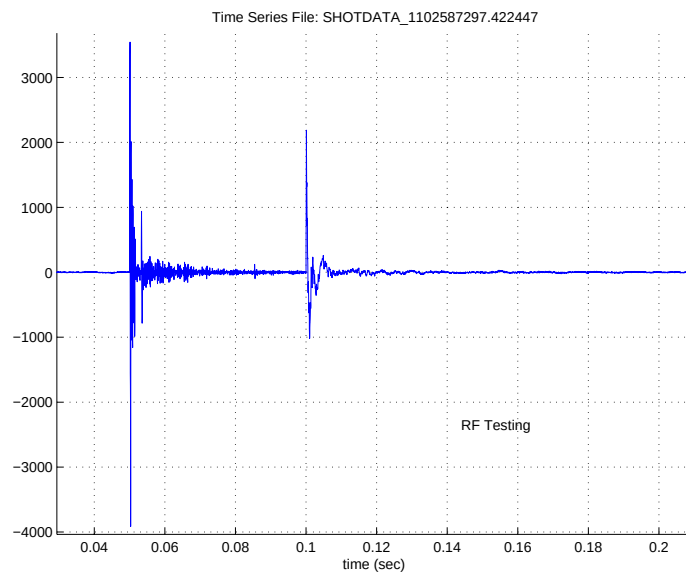


Fig. 12. Boomerang II shot data in the presence of high power RF transmissions.

1.2.2.4. Signal processing unit

The interior layout of the electronic box was greatly simplified due to the elimination of a number of components. Since the sensor signal Analog-to-Digital conversion was re-located to the DigiHub, the original PC-100 A/D daughter card inside the SPU was eliminated, along with the analog TDOA board. The WiFi and data FLASH memory daughter boards were also removed in favor of an external WiFi bridge option and USB FLASH memory, respectively.

1.2.2.5. Integrated custom display

An updated list of display requirements was compiled through meetings and correspondence with DARPA and the Boomerang user community. The list below includes new and original display features:

- 12 green LED compass rosette
- LED dimmer control
- System power on/off switch
- Audio output mute control
- Onboard audio output speaker
- Gunner headphone jack
- GPS coordinate alphanumeric display
- Multimodal shooter location alphanumeric display
- Control button for multimodal shooter location alphanumeric output button
- Similar or reduced cabling complexity vs. original Boomerang I design
- Reduced cost (< \$350)
- Low component lead time
- Weeks-scale design & implementation schedule

Subsequent discussions with the vendor of the original COTS display established that they could not provide a suitable set of desired new features within the design cost and time limits. The solution was to completely redesign and re-implement the display, making maximal use of COTS subcomponents. The new design accommodates the entire updated requirements list. In addition, GPS and Ethernet jacks are provided into which a GPS receiver and a maintenance computer unit is/can be plugged, respectively. The new display is built into a weatherized, electromagnetically shielded enclosure measuring five inches on a side and three inches deep as shown in Fig. 13.



Fig. 13. Re-designed Boomerang II display.

The TEST and MODE buttons of the display Keypad are used to select among three operational modes for Boomerang. Two of these are “normal operating modes,, and are identical except for the Alphanumeric Display format. Pressing and releasing the MODE button cycles Boomerang between the two normal operating modes. The third mode is “whistle test mode,,, which is toggled by pressing and releasing the TEST button.

The display and its controls operate as follows in either of the two normal operating modes:

When the system is initially powered up via the System Power Switch, the Alphanumeric Display indicates a status of “BOOTING,,. After booting has completed, the Alphanumeric Display is organized as:

```
VEHICLE.12345678
RG 123m EL 123u MUT
```

Here, grid coordinates are displayed on the top line. Range (meters) and elevation (feet or meters, depending on operational mode, with *u* being replaced by *f* or *m*, respectively) to the shooter are displayed on the bottom line. MUT is displayed when the aural output is toggled off via the MUTE key on the keypad. While no shot has been detected, all shot-related data are displayed as “-,, and the Compass Rosette remains dark. If no GPS lock is available, then “NO GPS,, is displayed in place of the grid coordinate data. A “,, after the word “VEHICLE,, blinks approximately once every two seconds to indicate overall good health of the Boomerang system.

When incident weapons fire is detected, the Compass Rosette illuminates the relative azimuth and the Alphanumeric Display indicates the range and elevation to the shooter as shown in Fig. 14. The display remains in this state for thirty seconds or until another shot is detected, whichever is sooner. A corresponding announcement is also made via the loudspeaker: “Shot *x*-o’clock!,, where *x* is 1 through 12.



Fig. 14. Re-designed Boomerang II display reporting shooter location.

The whistle test mode is provided to verify the operational health of all Boomerang’s microphones and microphone electronics. When this mode is first selected, the hub numbers of the microphones are displayed on the top line of the Alphanumeric Display. Seven “-,,s are displayed on the bottom line, one beneath each microphone channel:

```
MICROPHONE 1234567
HEALTH -----
```

Boomerang then listens for a standard “drill-instructor,, style whistle, typically carried by military personnel, to sound near the sensor array. When the whistle is blown, Boomerang indicates on which individual microphone(s) it has been detected by replacing the corresponding “-, with a “+,,. For a properly functioning system, every microphone channel will detect the sound:

```
MICROPHONE 1234567
HEALTH      +++++++
```

If any “-, fails to become a “+,, then the corresponding microphone is bad and should be replaced.

2. SYSTEM PERFORMANCE

2.1. Key technologies

Installation and operational requirements demand that there be but one microphone array. Shooter localization can be reliably done with just one array when both the bullet shockwave and the rifle muzzle blast are detected. Essentially, one needs only to measure the incoming direction of these two signals and the time difference between their arrivals. From the shockwave parameters, the caliber is found and an accurate ballistics model is used to complete the solution.

Urban environments and vehicle motion and combat contribute to unpredictably large noises that occupy the same frequency band as the muzzle blast signal making detection of the muzzle blast problematic, especially for distant shooters. To deal with this problem, Boomerang detection algorithms operate in a hierarchy according to the quality of the putative muzzle blast signal. When the muzzle blast signal is detected in the full bandwidth signal stream, the time difference of arrivals (TDOA) between them are found by cross correlation and used in a least-squares plane-wave incidence angle finder. When the residuals are sufficiently small, then the muzzle blast parameters and the shockwave parameters and ballistics model altogether produce the solution that is reported.

When the muzzle blast signal is not detected, a search is made in the filtered signal streams for peaks in power. The TDOA's between them are found by cross correlation as before. The relatively narrow band of the muzzle filter band reduces the resolution of these TDOA's. The incoming angle and arrival time computed from these TDOA's are less accurate than those computed from full bandwidth data (called a 'coarse' solution). The shooter solver then works as before, but with less accurate data.

When the muzzle blast signal is not detected, or when the angle of its arrival cannot be determined (too high residual value), then the system performs a shock-only solution. A genetic algorithm attempts to measure the incoming shock angle, its curvature, and the gradient of curvature. When all three aspects of the measurement are done correctly, the algorithm finds the bullet trajectory. The location of the shooter along the trajectory cannot be found, but the shooters relative bearing and elevation are reasonably accurately computed from an assumed distance. When just the first two aspects of the incoming shock cone are found correctly, then the muzzle detection file is consulted to find the muzzle blast angles with least residuals. Essentially, we need just to distinguish between two angles that are about 120 degrees apart. This approach allows the correct shooting direction to be found most times. When these angles do not distinguish between the two possible directions, both aspects (both ends of the trajectory) and elevations to the shooter are reported. The one with the lowest residual is reported first.

2.2. Test results

2.2.1. Reliability testing

A series of environmental tests (i.e. EMI, vibration, and thermal) were conducted at IQS Marlboro Test Labs to characterize the reliability of the system operating over the range of expected operational conditions. EMI testing was conducted in accordance with Mil-461-E to verify low emissions and low susceptibility to RF interference. Vibration testing was conducted in accordance with Mil-810C. Thermal testing was conducted to validate performance at 50

degrees C. Preliminary thermal testing demonstrated the need for a fan internal to the signal processing enclosure to cool various components on the CPU. After successful testing at 50 degrees C with the fan installed, thermal and voltage extreme testing, known as 4-corner testing, was conducted to verify system performance at temperature extremes (-13 deg C to +50 deg C) and voltages extremes (20 to 28 VDC).

In addition to the environmental tests conducted at IQS Marlboro Test Labs, the US Army Aberdeen Proving Grounds performed RF emissions and susceptibility testing and a 500-mile road test over varied terrain to test structural integrity. This testing demonstrated that Boomerang II was compatible with SINCGARS radio and remained operational after the 500-mile road test.

2.2.2. Performance test results

Boomerang has been exhaustively tested under a number of 'live fire' tests. BBN has conducted ten tests at the Devens, MA Reserve Forces Training Area to evaluate system hardware components and software improvements. The US Marine Corps Warfighting Laboratory (MCWL) at Quantico, VA has coordinated four tests including a Limited Technical Assessment (LTA) in January 2004. This LTA evaluated Boomerang performance in an open field and a highly reverberant urban (MOUT facility) environment. These tests were conducted static and moving using a remote-controlled ATV as shown in Fig. 15. Deployment of Boomerang I to support OIF II was based on the successful performance demonstrated during the US Marines LTA.



Fig. 15. Boomerang I mobile testing on a remote-controlled ATV.

More recently, the US Army Aberdeen Proving Grounds conducted a series of tests in January/February 2005 on Boomerang II to evaluate RF compatibility, road testing and subsequent 'live fire' system performance. The 'live fire' tests were conducted in an open field at the completion of the 500-mile road test. Static tests were conducted by firing various small arms weapons at several shooter distances and CPA's//Boomerang (1 meter & 30 meters). Various vehicle orientations were also tested (0° , 90° and 120°). This series was repeated in a mobile test using a remote-controlled HMMWV as shown in Fig. 16. The white streak over the top of the vehicle is a tracer round being fired behind the HMMWV while being remotely driven at 45 MPH.



Fig. 16. Boomerang II mobile testing on a remote-controlled HMMWV.

The consistency in reported shooter azimuth is shown in Fig. 17 and Fig. 18 for the static test series at 100 meter and 300 meter shooter distances.

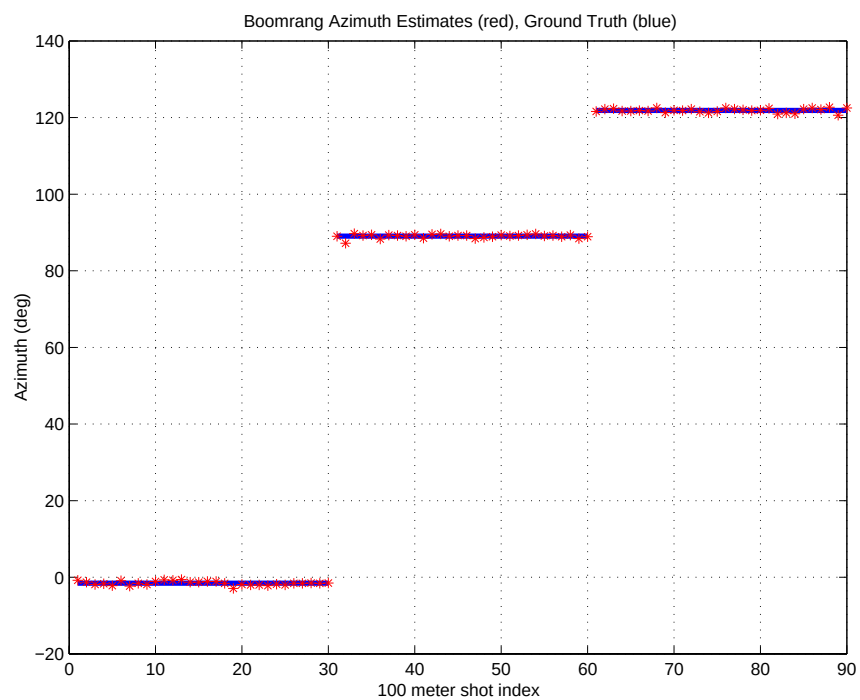


Fig. 17. Boomerang II static azimuth performance at 100-meter shooter distance.

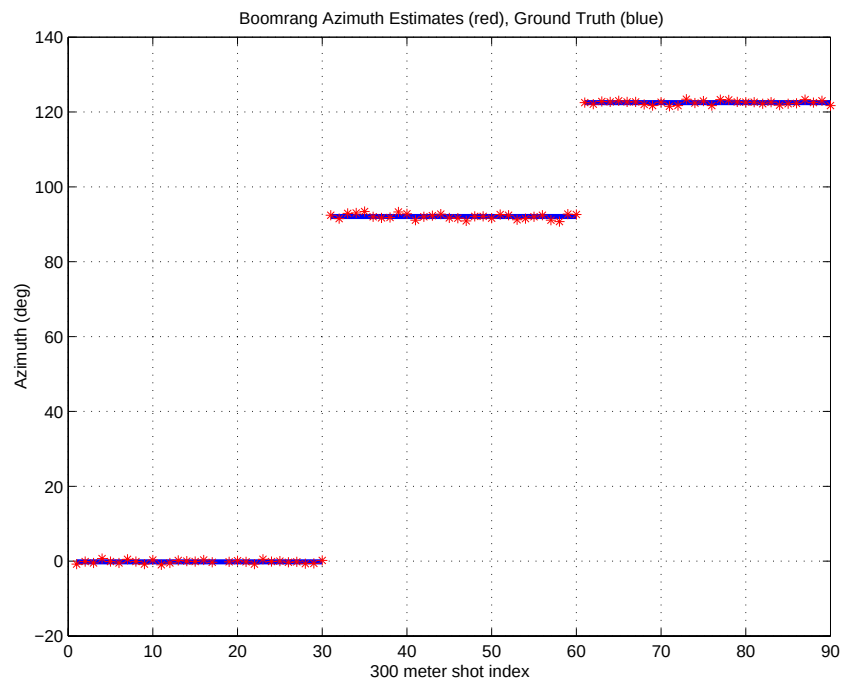


Fig. 18. Boomerang II static azimuth performance at 300-meter shooter distance.

During the mobile test series, Boomerang II arrays were mounted on the driver side and passenger side to simultaneously evaluate system performance at the two primary array installation sites. The composite performance (static and mobile) of Boomerang for all shots at a 100-meter shooter distance is shown in Fig. 19. A slight degradation in performance occurs for the mobile condition due to the increased noise levels from the HMMWV operating at 45 MPH. An ‘ambiguous’ solution is produced when the muzzle blast cannot be identified. Under this condition, the solver attempts to compute a solution based on the shock wave that results in two solutions due to the inability to sufficiently resolve shock cone parameters.

Boomerang Performance @ 100 m (All Shots)

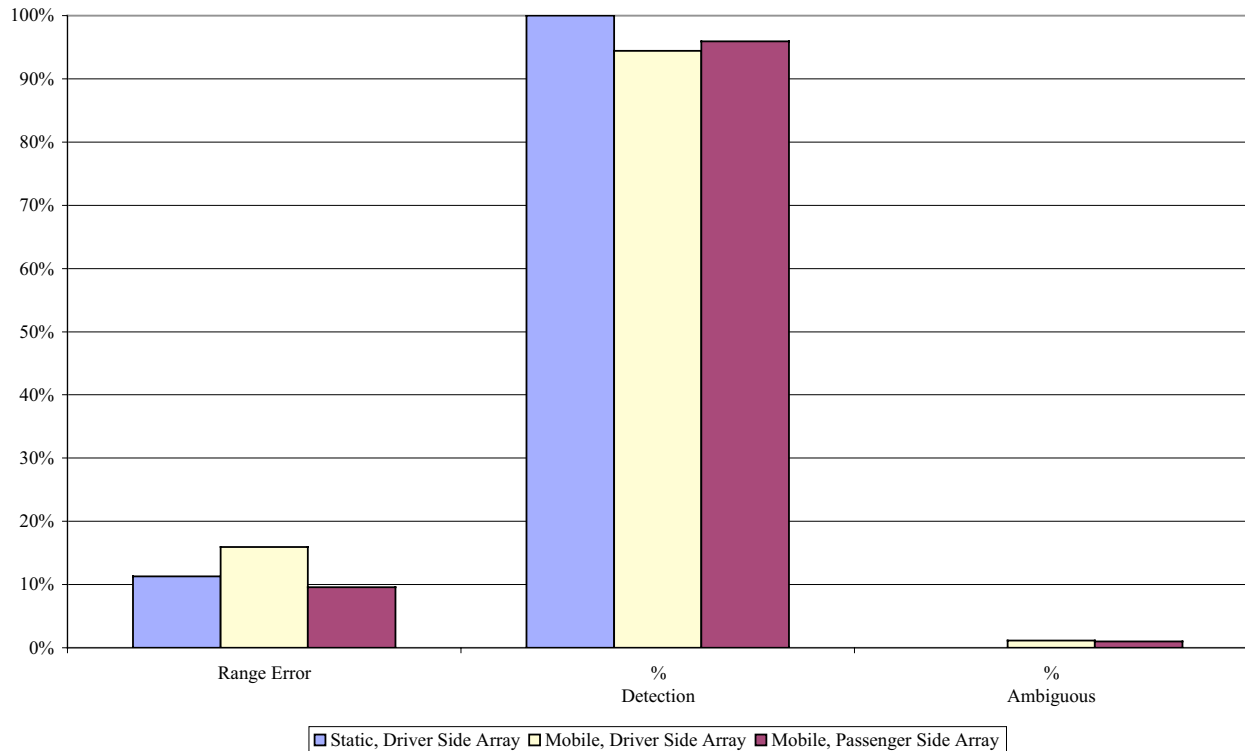


Fig. 19. Boomerang II composite performance at 100-meter shooter distance.

A summary of Aberdeen Proving Grounds test results for the static and mobile test conditions is presented in Tables 1 and 2. These tables show a slightly increasing range error with increasing range and a remarkably small standard deviation in azimuth and elevation (static case). It should be noted that the larger standard deviation in azimuth for the moving case is due primarily to the variability in ground truth caused by firing at slightly different times while the vehicle is traveling at a high speed. There were no reported false alarms during static and mobile testing.

Table 1. Boomerang II Test Results – Static, Aberdeen Proving Grounds

Test Condition	Range Error	Range Std Dev	Azimuth Std Dev	Elevation Std Dev	Total Shots	% Detection	% Ambiguous
AVERAGE (All Shots) @ 100 m	11%	12	1.5	0.8	180	100.0%	0.0%
AVERAGE (All Shots) @ 300 m	31%	51	0.7	1.2	180	99.4%	0.0%

Table 2. Boomerang II Test Results – Mobile (45 MPH), Aberdeen Proving Grounds

Test Condition	Range Error	Range Std Dev	Azimuth Std Dev	Elevation Std Dev	Total Shots	% Detection	% Ambiguous
AVERAGE (All Shots) @ 100 m	16%	22	11.0	3.7	180	94.4%	1.1%
AVERAGE (All Shots) @ 200 m	30%	67	11.3	6.0	30	96.7%	0.0%

2.3. Boomerang II performance features and specifications

Features

- Operates on moving vehicles up to 60 mph, all terrains and environments (open field/urban)
- Detects shooter on first round fired under all environmental conditions (day/night/fog/rain/snow/sandstorm)
- User Outputs: Visual/Aural. Custom LED display of relative shooter azimuth, alphanumeric display, and integrated speaker. Vehicle military coordinate displayed in standby mode.
- RFI compatibility: Meets MIL-STD-461E
- Simple Operation: On/Off switch, no calibration requirements
- Built-in system diagnostics and Ethernet interface
- Power Requirements: 24 VDC, 25 Watts

Performance

- Probability of Detection: > 99% detection of supersonic projectiles (.20 to .50 cal)
- Detection Accuracy: < 2.5 deg error (rms), 95% of detected shots within 4 degrees
- Response Time: Less than 1 second
- False Alarms: None due to wind noise, bumps, door slams
- Return Fire Suppression - No alert from outgoing fire

Cost

- Less than \$10k (USD) per system

3. CONCLUSION

Boomerang II is a reliable and accurate shooter detection system designed to operate statically or on a moving vehicle under a wide variety of operational conditions. The system has been designed to detect supersonic projectiles fired in close proximity to the sensor array and provide an immediate indication (aural and visual) of the relative shooter direction, range and elevation. Significant efforts have been implemented to prevent system false alarms caused by non-ballistic events such as road bumps, door slams, wind noise and extraneous noise events (vehicle traffic, firecrackers, urban activity). The system will not alert when shots are fired from the vehicle (return fire). The Boomerang design philosophy has been to keep system operation and installation as simple as possible. Boomerang II has been designed for compatibility with military tactical radio systems and will operate under a wide range of environmental conditions. The system unit cost is sufficiently low that it can be installed and operated on a large number of US military vehicles. Boomerang will not stop a speeding bullet, however, it will alert our troops when they have entered into 'harm's way' and give them a fighting chance.

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